

FLINT WATER LINE PROJECT

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Project Overview

Flint Water Crisis

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FLINT WATER CRISIS

In 2014, Flint made the switch to have their water source be the Flint River. This led to the leaching of lead into the water because of old corroded service lines.

Dangerous amounts of lead were in the water leading to many health concerns.

The city began replacing the water lines slowly, with limited funding.

OUR GOAL

The limited funding creates a need for precise digging and replacing of the pipes.

Our goal was to piece a model together to find the best prediction model resulting in high accuracy location of lead pipes in the city.

This will allow for the funding to be used in the most beneficial and efficient way.

Predictor Selection

Predictors

Predictor Selection

To determine the most influential predictors of dangerous service lines, we explored multiple methods:

- Stepwise Logistic Regression: Initially used on continuous variables; however, challenges arose when combining nominal and continuous data types.
- CART (Classification and Regression Trees): We ran CART models separately for continuous and nominal predictors, then combined the results. CART provided useful column contribution scores to identify key features.
- JMP Predictor Screening Tool: This tool ultimately offered the most intuitive and consistent measure of variable importance across all types. It produced a ranked distribution of predictor contributions, which we cross-referenced with CART results.

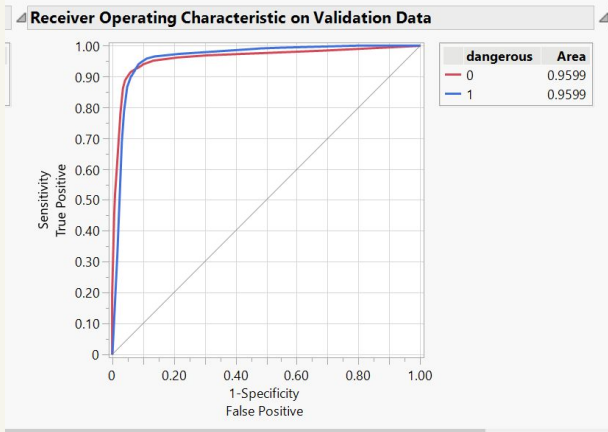
Based on this combined approach, we selected the following top 10 predictors:

- Year Built: *Older homes more likely to have lead*
- SL_Type: *Historical service line type*
- Parcel Acres: *Larger parcels often newer*
- Land Value: *Potential proxy for infrastructure quality*
- HomeSEV: *State estimated property value*
- CENTRACT: *Census tract—spatial clustering observed*
- Residential Building Value: *High value homes may correlate with pipe upgrades*
- PRECINCT: *Political zone potentially related to replacement history*
- Hydrant Type: *Infrastructure variable (e.g., age of hydrants)*
- Last_Test: *Latest water test lead level*

Models Tried

CART, Bootstrap
Neural Networks, Regression

Models Tried

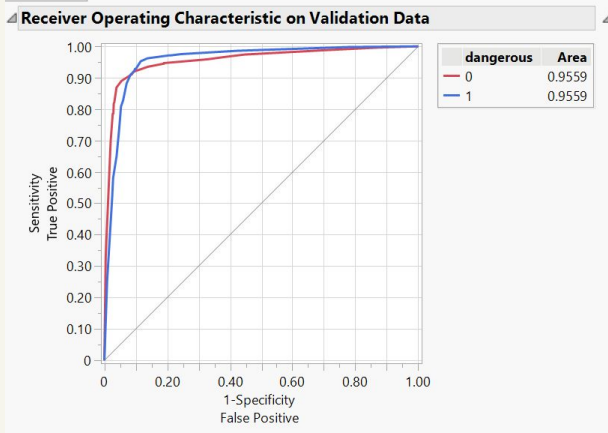


Decision Tree
Model (CART)

AUC: 0.9599

Predictors Used

- Year Built
- Service Line Type (SL_Type)
- Precinct
- Hydrant Type
- Census Tract (Centract)
- Parcel Acres



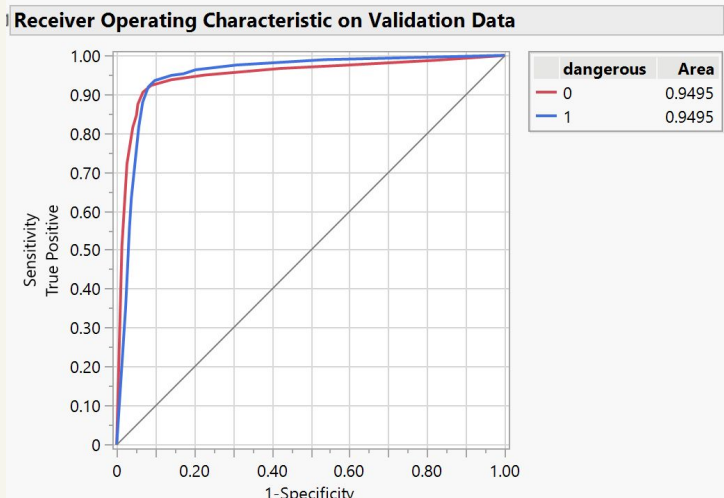
Neural Network

AUC: 0.9559

Predictors Used

- Year Built
- Service Line Type (SL_Type)
- Precinct
- Hydrant Type
- Census Tract (Centract)

Models Tried



Bootstrap

AUC: 0.9495

Predictors Used

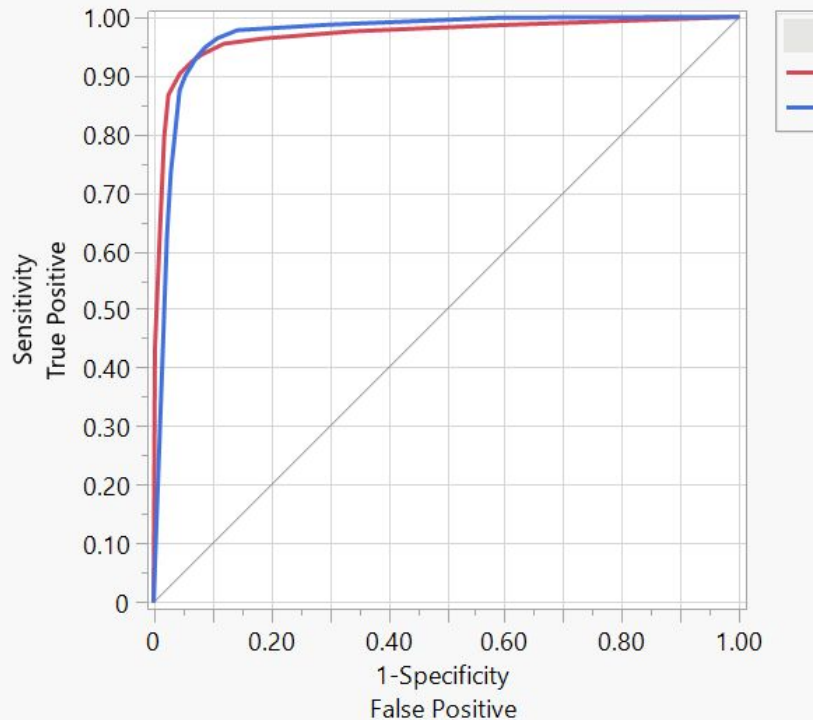
- Year Built
- Service Line Type (SL_Type)
- Precinct
- Hydrant Type
- Census Tract (Centract)
- Parcel Acres

Final Model

Boosted Tree

Boosted Tree Model

Our final predictive model was a **Boosted Tree** with a strong **AUC of 0.9692**, indicating excellent classification performance. The model prioritized five key predictors: **Census Tract (CENTRACT)**, **Hydrant Type**, **Year Built**, **Service Line Type (SL_Type)**, and **Precinct**. These variables provided valuable signals about infrastructure age, geographic risk, and historical installation patterns. Notably, **CENTRACT** and **Hydrant Type** contributed the most, highlighting the importance of **location-based and utility-specific features** in identifying high-risk service lines.



Actual	Predicted Rate	
	0	1
0	0.931	0.069
1	0.074	0.926

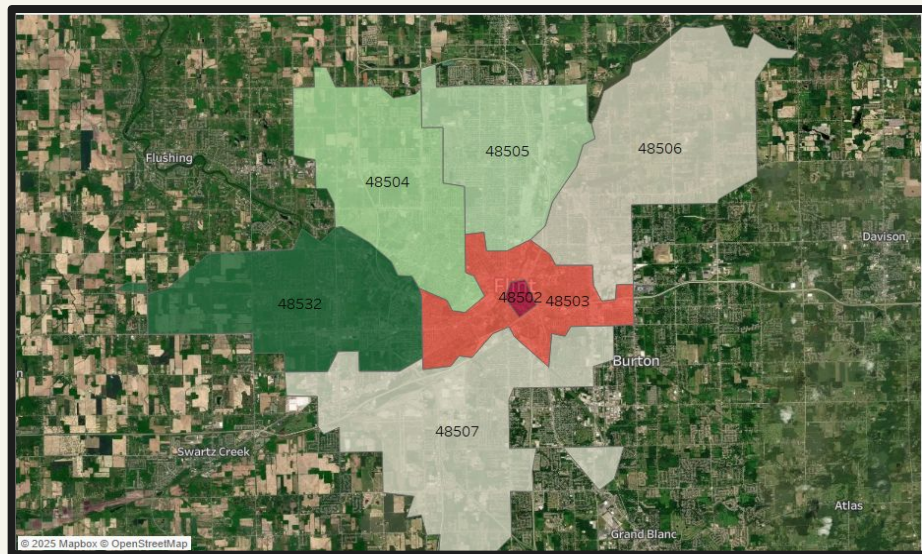
	dangerous	Area
0		0.9692
1		0.9692

Results & Conclusion

The Bottom Line

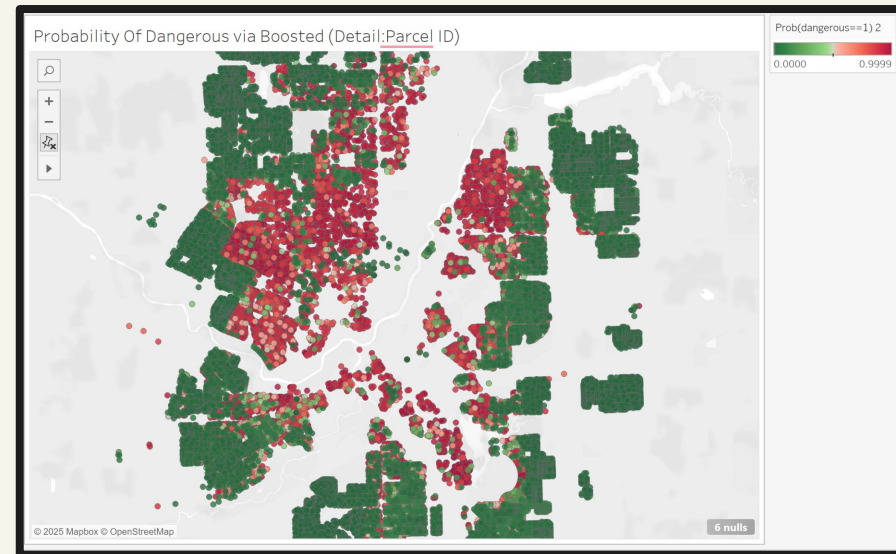
Maps of High Risk Areas

BY ZIPCODE



 - Low Risk
 -High Risk

BY PARCEL ID



Results & Conclusion

Final Model: Boosted Tree

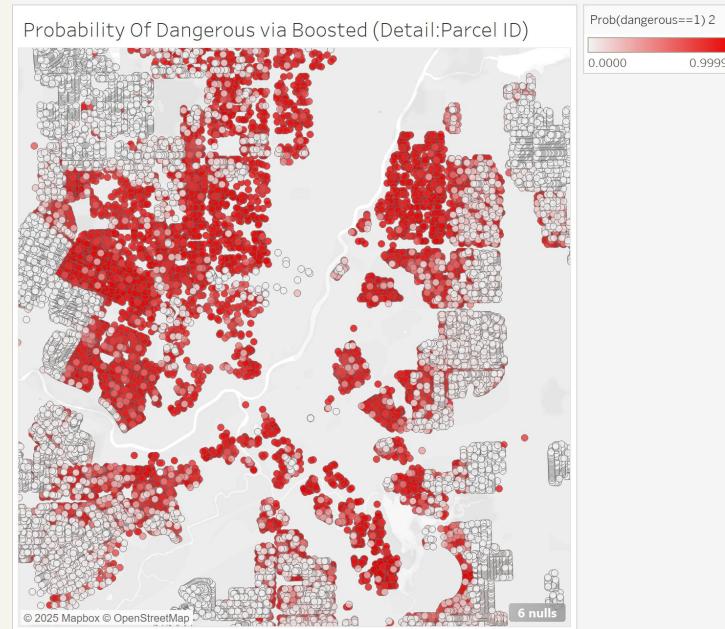
AUC (Validation): 0.9692 — Excellent classification performance

AUC (Training): 0.986 → Gap of **0.017** shows minimal overfitting

Misclassification Rate: 0.0709- Very low across both training and validation sets

High predicted danger scores were **concentrated in older neighborhoods**

Visual patterns matched known high-risk areas, reinforcing model reliability



Creator	.2 .4 .6 .8	Entropy RSquare	Generalized RSquare	Mean -Log p	RASE	Mean Abs Dev	Misclassification Rate	N
Model Averaged		0.2507	0.3893	0.5119	0.3759	0.2126	0.1674	26863
Neural Model NTanH(3) - Save Formulas		0.2090	0.3334	0.5404	0.3870	0.2150	0.1789	26863
Partition		0.1978	0.3179	0.548	0.3795	0.1991	0.1663	26863
Bootstrap Forest		0.1738	0.2838	0.5644	0.3870	0.2237	0.1778	26863
Boosted Tree		0.1445	0.2405	0.5845	0.3697	0.1869	0.1573	26863

Questions ?